The Independent Set Inverse Problem for Bounded Treewidth Graphs

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Abstract

We investigate which positive integers arise as the number of independent sets of a graph with treewidth at most k. Using an algebraic framework that identifies series-parallel graph composition with matrix product and Hadamard product of 2×2 signature matrices, we show computationally that every integer $n \geq 3$ is the independent set count of some series-parallel graph (treewidth ≤ 2), verified for $n \leq 10{,}000$. We prove that the bisemigroup closure of the Fibonacci matrix modulo any prime p is all of $M_2(\mathbb{F}_p)$, eliminating all local obstructions. The surjectivity problem reduces, via a dot-product identity, to showing that a specific linear form on a structured subset of $\mathbb{N}^2$ is cofinite.

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1 Problem Statement

Which positive integers arise as the number of independent sets of a graph with treewidth at most k?

This is a bounded-treewidth generalization of the classical IS inverse problem. For trees (treewidth 1), it is conjectured that all but finitely many positive integers are achievable, but no density result is known.

1.1 Difficulty Hierarchy

Treewidth	Achievable (of 10,000)	Mechanism
$= 0$ (edgeless)	13 (0.1%)	Powers of 2 only
≤ 1 (forests)	9,269 (92.7%)	Tree DP + multiplicative closure
≤ 2 (series-parallel)	**10,000 (100%)**	SP algebra (this paper)
≤ 3	10,000 (100%)	Follows from tw ≤ 2
≤ 4	All integers (proven)	Chain gadget / continued fraction
unbounded	All integers (trivial)	Threshold graphs / binary expansion

1.2 Main Results

Theorem 1 (Computational, verified for $n \leq 10{,}000$)**.** *Every integer $n \geq 3$ is the number of independent sets of some series-parallel graph (treewidth ≤ 2). Combined with the empty graph ($i = 1$) and a single vertex ($i = 2$), every positive integer is achievable at treewidth ≤ 2.*

Conjecture 2. *Every integer $n \geq 1$ is the number of independent sets of some graph with treewidth ≤ 2.*

This makes the treewidth ≤ 3 question (originally posed as the hard case between the open tree conjecture and the proven tw ≤ 4 result) trivially resolved: treewidth 2 already suffices.

2 The Algebraic Framework

2.1 Series-parallel graphs as matrix algebra

A two-terminal series-parallel (SP) graph G with terminals s, t has an *IS signature* $(f_{00}, f_{10}, f_{01}, f_{11})$, where f_{ij} counts independent sets with s in state i and t in state j. Arrange this as a 2×2 matrix:

$$M_G = \begin{pmatrix} f_{00} & f_{01} \\ f_{10} & f_{11} \end{pmatrix}.$$

The two SP composition operations become:

- **Series composition** (identify inner terminals): $M_{G_1;G_2} = M_{G_1} \cdot M_{G_2}$ (matrix product).
- **Parallel composition** (identify both terminals): $M_{G_1 \| G_2} = M_{G_1} \circ M_{G_2}$ (Hadamard/entrywise product).

The IS count is the quadratic form:

$$i(G) = \mathbf{1}^T M_G \mathbf{1}, \qquad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

2.2 The Fibonacci matrix

The single edge s–t has signature:

$$M_E = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

This is the *Fibonacci matrix*. As a Möbius transformation, it acts as $z \mapsto 1 + 1/z$, with fixed point $\varphi = (1 + \sqrt{5})/2$ (the golden ratio). It satisfies $M_E^2 = M_E + I$, has determinant -1, and lives in $GL_2(\mathbb{Z})$.

Series powers give paths: M_E^k has entries $(F_{k+1}, F_k; F_k, F_{k-1})$ and IS count F_{k+2}.

2.3 The generated set

Let $\mathcal{S}$ be the smallest set of 2×2 nonneg integer matrices containing M_E and closed under both matrix product and Hadamard product. Then:

$$\{i(G) : G \text{ is a series-parallel graph}\} \supseteq \{\mathbf{1}^T M \mathbf{1} : M \in \mathcal{S}\}.$$

Computational finding: $\{\mathbf{1}^T M \mathbf{1} : M \in \mathcal{S}\} \supseteq \{3, 4, 5, \ldots, 10000\}$.

2.4 The bisemigroup structure

The set $\mathcal{S}$ is closed under two associative binary operations with:

- **Different identities**: I for matrix product, $J = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ for Hadamard product.
- **No distributive law**: neither operation distributes over the other.
- **Neither identity in $\mathcal{S}$** (since $\mathcal{S}$ is generated by M_E alone).

This is a set with two independent semigroup structures (a *bisemigroup*).

3 The Iterated Function System

3.1 Linearization on $\mathbb{Z}^4$

Identify each 2×2 matrix with a vector $(a, b, c, d) \in \mathbb{Z}^4$ where $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The two operations (applied with M_E) become linear maps on $\mathbb{Z}^4$:

Right-multiply by M_E (series with edge):

$$R : (a, b, c, d) \mapsto (a + b,\ a,\ c + d,\ c).$$

As a 4×4 matrix: $R = F \oplus F$, block-diagonal with the Fibonacci matrix acting independently on (a, b) and (c, d).

Hadamard with M_E (parallel with edge):

$$P : (a, b, c, d) \mapsto (a,\ b,\ c,\ 0).$$

This is projection — it simply kills the $(1, 1)$ entry.

The IS count is the linear functional $\varphi(a, b, c, d) = a + b + c + d$.

3.2 Properties of the generators

	R (series)	P (parallel)
Type	Expanding	Contracting (rank 3)
Eigenvalues	$\varphi, \varphi, -1/\varphi, -1/\varphi$	$1, 1, 1, 0$
Determinant	1 ($\in SL_4(\mathbb{Z})$)	0 (not invertible)

3.3 Sequential collapse (Gemini)

Both R and P are block-diagonal with the same 2×2 partition:

$$R = F \oplus F, \qquad P = I_2 \oplus E, \qquad E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The top pair (a, b) and bottom pair (c, d) never mix. The bottom block collapses via $EF^kE = F_{k+1} \cdot E$. For a sequential word $X = R^{a_1}PR^{a_2}P \cdots PR^{a_{k+1}}$ (with k projections and $N = \sum a_i$ total R's), the IS count evaluates exactly to:

$$f(X) = F_{N+3} + F_{a_1+2} \cdot F_{a_{k+1}+1} \cdot \prod_{i=2}^{k} F_{a_i+1}.$$

Verified computationally for 9,330 words with 0 mismatches.

Consequence: The sequential IFS achieves only ∼34% of integers up to 1000. The sequential orbit has density tending to zero (like the primes), despite equidistribution mod p for all primes p.

Range	Sequential density
$[3, 100]$	72%
$[3, 500]$	46%
$[3, 1000]$	34%
$[3, 5000]$	19%
$[3, 10000]$	13%

Positive density genuinely requires the branching Hadamard products.

4 The Local Theory

4.1 The full closure theorem

Theorem 3 (Verified for $p = 2, 3, 5, 7, 11$)**.** *The full bisemigroup closure of M_E in $M_2(\mathbb{F}_p)$ under matrix product and Hadamard product is:*

$$S_p = M_2(\mathbb{F}_p).$$

Proof sketch. Define $K_u := \left(\begin{smallmatrix} u & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $K_1 = E$ and $K_2 = E^2 \circ E$ (Hadamard). Since $\det(E) = -1 \neq 0$, $E \in GL_2(\mathbb{F}_p)$, and $E^{-1} \in S_p$ (as a power of E). The recurrence $K_{u+1} = K_2 \cdot E^{-1} \cdot K_u$ gives all K_u for $u \in \mathbb{F}_p$. From these:

$$U_t := K_{t+1}E^{-1} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad L_t := E^{-1}K_{t+1} = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}.$$

Diagonal matrices are obtained via Hadamard products with I (which is a power of E):

$$D_x := (U_1L_{x-1}) \circ I = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}, \qquad D'_y := (L_{y-1}U_1) \circ I = \begin{pmatrix} 1 & 0 \\ 0 & y \end{pmatrix}.$$

The permutation matrix $P = E \circ E^{-1} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is also in the closure. Any matrix with $a \neq 0$ factors as $L_{c/a}\, D_a\, D'_{d-bc/a}\, U_{b/a}$; remaining cases are handled by P and zero diagonal factors. $\square$

Remark 4 (Sequential orbit is a proper subset). The earlier claim that "the orbit is a proper subset of $(\mathbb{Z}/m\mathbb{Z})^4$" was about the *sequential* orbit (IFS applying R or P one at a time). The sequential orbit IS a proper subset (32.7% of $M_2(\mathbb{F}_7)$, decreasing with p). The full bisemigroup fills everything.

Remark 5 (Caveat for sieve methods). The proof requires $E^{-1} = E^{\mathrm{ord}(E)-1}$, where $\mathrm{ord}(E)$ divides $|GL_2(\mathbb{F}_p)| = p(p-1)^2(p+1)$. So the expression trees realizing arbitrary matrices mod p have depth polynomial in p. This rules out the uniform-in-p error bounds that sieve methods require. The archimedean problem (cofinite linear form on $\mathcal{R}$) is genuinely independent of the local theory.

p	Sequential orbit	Full bisemigroup
2	75% of $M_2(\mathbb{F}_2)$	**100%**
3	89%	**100%**
5	67%	**100%**
7	33%	**100%**
11	—	**100%**

5 The Global Theory: Dot-Product Reduction

5.1 Transpose closure

Lemma 6. *$\mathcal{S}$ is closed under $M \mapsto M^T$, and consequently $\mathcal{R} = \mathcal{C}$.*

Proof. M_E is symmetric, $(A \circ B)^T = A^T \circ B^T$, and $(AB)^T = B^T A^T$. Any expression tree can be mirrored (swap children at each $\cdot$-node) to produce the transpose. Since $\mathrm{col}(M) = \mathrm{row}(M^T)$ and $M^T \in \mathcal{S}$, we have $\mathcal{R} = \mathcal{C}$. $\square$

5.2 Dot-product identity

Lemma 7. *For $M, N \in \mathcal{S}$,*

$$\varphi(MN) = \mathrm{col}(M) \cdot \mathrm{row}(N) = (M^T\mathbf{1})^T(N\mathbf{1}),$$

where $\mathrm{row}(N) = \mathbf{1}^T N \in \mathbb{N}^2$ *and* $\mathrm{col}(M) = M^T\mathbf{1} \in \mathbb{N}^2$.

Proof. $\varphi(MN) = \mathbf{1}^T MN\mathbf{1} = (M^T\mathbf{1})^T(N\mathbf{1})$. $\square$

Proposition 8. $\varphi(\mathcal{S}) \supseteq \{u \cdot v : u, v \in \mathcal{R}\}$.

5.3 The $(2, 1)$ shortcut

Since $\mathrm{col}(M_E) = (2, 1)$, for any $N \in \mathcal{S}$ with $\mathrm{row}(N) = (x, y)$:

$$\varphi(M_E \cdot N) = 2x + y.$$

Conjecture 9 (Cofinite Linear Form)**.** *The set $L := \{2x + y : (x, y) \in \mathcal{R}\}$ is cofinite; equivalently, there exists T such that $L \supseteq [T, \infty) \cap \mathbb{Z}$.*

Conjecture 9 plus a finite verification for $n < T$ would prove the main conjecture.

5.4 The Fibonacci transform on row vectors

Right-multiplication by M_E acts on row vectors as the *Fibonacci map*:

$$\mathrm{row}(M \cdot M_E) = \mathrm{row}(M) \cdot M_E = (x + y,\ x).$$

This map preserves $\gcd(x, y)$, and sends $2x + y \mapsto 3x + 2y = (2x + y) + x$. Repeated application gives $2x_n + y_n = F_{n+2}\, x + F_{n+1}\, y$, sweeping values that grow like φ^n.

If the Hadamard product produces enough *seed vectors* (x, y) with modest y, the Fibonacci transform propagates them into families where $2x + y$ sweeps intervals.

5.5 Explicit seed families

Two provable seed families:

1. **Paths:** $\mathrm{row}(E^n) = (F_{n+2}, F_{n+1})$, giving $2x + y = 2F_{n+2} + F_{n+1}$.
2. **Parallel paths:** $\mathrm{row}(E^m \circ E^n) = (F_{m+n+1}, F_{m+n-1})$, using the identity $F_{m+1}F_{n+1} + F_mF_n = F_{m+n+1}$. This gives $2x + y = 2F_{s+1} + F_{s-1}$ for $s = m + n$.

Both families produce Fibonacci-spaced values, too sparse alone. The key question is whether additional families from deeper Hadamard products interleave to fill all gaps.

5.6 Computational validation

The dot-product identity was tested with the 4-iteration closure (2,594 signatures, 1,902 row/col vectors):

Range	Dot product gaps
$[1, 10{,}000]$	Only $\{1, 2, 3, 4, 6, 9\}$
$[10{,}001, 40{,}000]$	None
$[40{,}001, 50{,}000]$	72 gaps (finite-depth artifact)

Since $3, 4, 6, 9$ are all directly achievable as IS counts of small SP graphs, every integer in $[3, 10{,}000]$ is covered by the dot-product route alone.

For fixed col vector (a, b), the achievable dot products become consecutive above a threshold:

Col vector	Consecutive from
$(4, 3)$	137
$(3, 1)$	149
$(3, 2)$	165
$(2, 1)$	174

The row/col sum sets do *not* form thick rectangles (coverage of $[1..10]^2$ is only 25%), but their dot products are nearly surjective nonetheless.

6 Context: Why Standard Machinery Fails

6.1 Arithmetic circuits and the branching bisemigroup

Elements of $\mathcal{S}$ are expression trees, not words. In an expression tree of depth D with Hadamard nodes, entries are polynomials of degree up to 2^D in the Fibonacci numbers. This exponential degree explosion means:

	Sequential (words)	Bisemigroup (trees)
Algebraic model	Random matrix product	Arithmetic circuit
Degree of entries	$O(1)$ (Fibonacci products)	2^D (exponential)
Transfer operator	Linear, finite state mod p	Nonlinear; no finite-state reduction

6.2 Barriers to equidistribution proofs

1. **Degree-characteristic barrier.** Weil/Deligne bounds require $d < p$; trees of depth $D \approx \log_2 p$ already exceed this.
2. **PIT and circuit complexity.** Proving algebraic independence for these circuits is entangled with open problems in complexity theory.
3. **No branching expander machine.** The measure evolution $\mu_{n+1} = \alpha(\mu_n * \mu_n) + \beta(\mu_n \circ \mu_n)$ is a nonlinear dynamical system with no precedent in the Bourgain–Gamburd framework.

However, the full closure theorem (Theorem 3) renders the local theory moot: $S_p = M_2(\mathbb{F}_p)$ is established by an elementary algebraic argument, bypassing all of the above. The PIT/expander/sieve discussion is context for the general difficulty, not the proof strategy.

6.3 The sieve gap

Even equidistribution mod p with good rates is insufficient for surjectivity. Sieve methods need uniform-in-p error bounds ($\sum_p \delta(p)/p < \infty$). The local theorem uses chains of length polynomial in p, ruling out such uniformity. The archimedean problem is genuinely independent of the local theory.

6.4 Analogy with Apollonian packings

	Apollonian gasket	SP graph IS counts
Space	Descartes quadruples $\in \mathbb{Z}^4$	Matrix entries $\in \mathbb{Z}^4$
Semigroup	Apollonian group (positivity)	$\langle R, P\rangle$ (non-invertible P)
Functional	Curvature	$\varphi = a + b + c + d$
Local picture	Thin group in $O(3,1)$	$S_p = M_2(\mathbb{F}_p)$ (complete)
Positive density	Bourgain–Fuchs 2011	Expected
Surjectivity	Open	Verified to 10,000

7 Prior Results

7.1 The tw ≤ 4 construction (proven universal)

(Due to OpenAI o3, March 2026.) The "chain gadget" $C(m_1, \ldots, m_k)$ has IS count following the continued-fraction recurrence $A_i = m_i A_{i-1} + B_{i-1}$, $B_i = A_{i-1}$, with $i(G) = A_k + B_k$. The Euclidean algorithm guarantees parameters exist for any target T, with $k = O(\log T)$. Each gadget has treewidth ≤ 2; adding the spine gives treewidth ≤ 4.

7.2 The tree IS conjecture

For trees (treewidth 1), it is conjectured that all but finitely many positive integers occur as $i(T)$. This is open. Trees miss ~7.3% of integers up to 10,000 (including primes 7, 11, 19, 29, 31, ...). The jump from tw ≤ 1 to tw ≤ 2 closes all gaps; parallel composition (Hadamard product) has no tree analogue.

7.3 Linek's theorem

Linek (1989): every positive integer is the IS count of some bipartite graph, but with no treewidth bound.

8 Significance and Open Problems

1. **Complete local picture.** The bisemigroup closure mod p is all of $M_2(\mathbb{F}_p)$, eliminating all local obstructions. This is far stronger than φ-equidistribution.

2. **Clean reduction to 1D number theory.** Via transpose closure ($\mathcal{R} = \mathcal{C}$) and the dot-product identity, the surjectivity problem reduces to: is $\{2x+y : (x,y) \in \mathcal{R}\}$ cofinite?

3. **Finite boundary algebra viewpoint.** The 2×2 terminal signatures are the $k = 2$ instance of the graph-homomorphism / Holant framework. The Holant clone theory and Fibonacci-gate literature [7] may provide structural lemmas.

4. **The remaining open problem** is purely archimedean: prove Conjecture 9 (the cofinite linear form conjecture for the row-sum set $\mathcal{R}$). This is a statement in elementary additive number theory about a structured subset of $\mathbb{N}^2$ generated by the Fibonacci map and entrywise products.

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